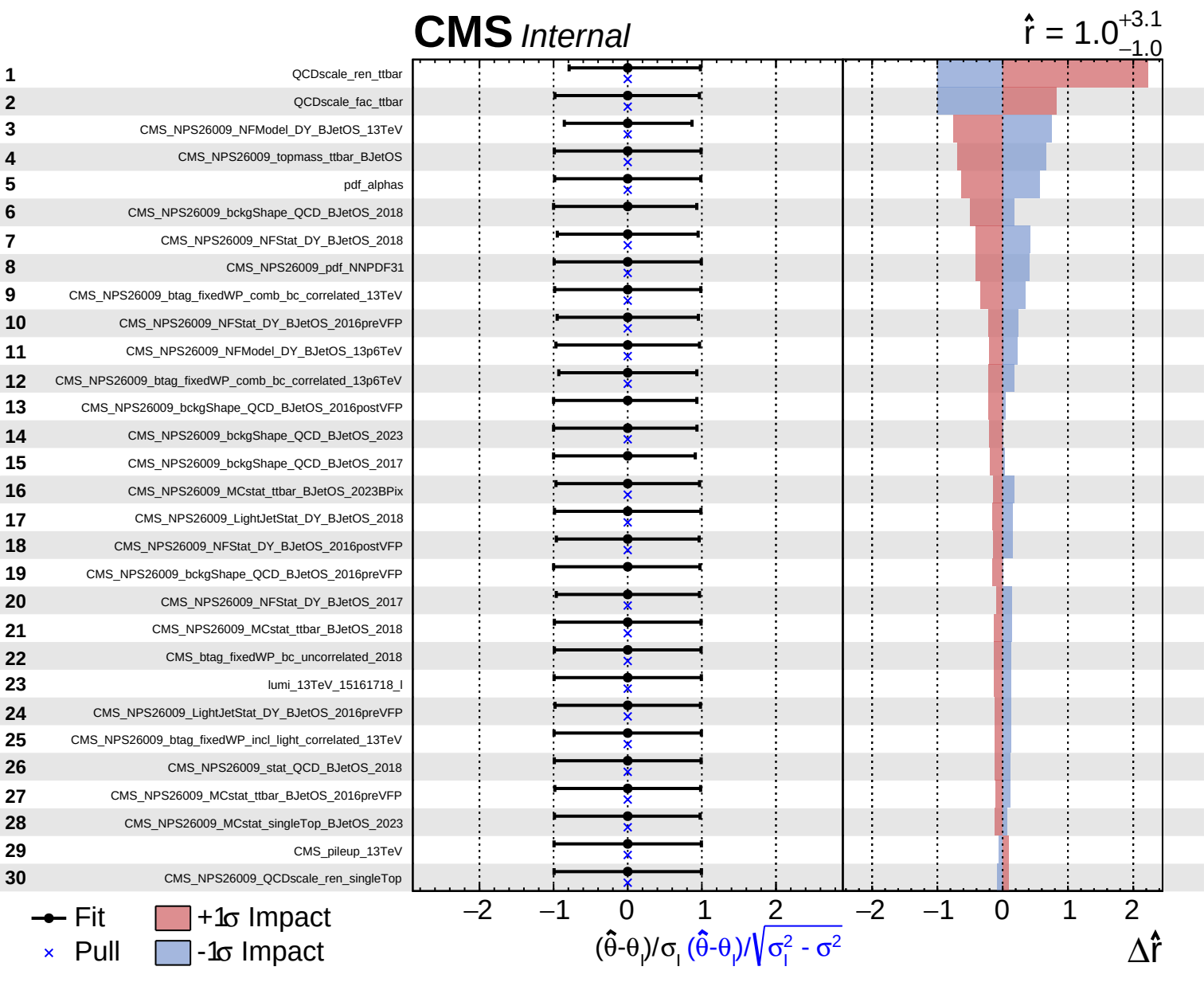
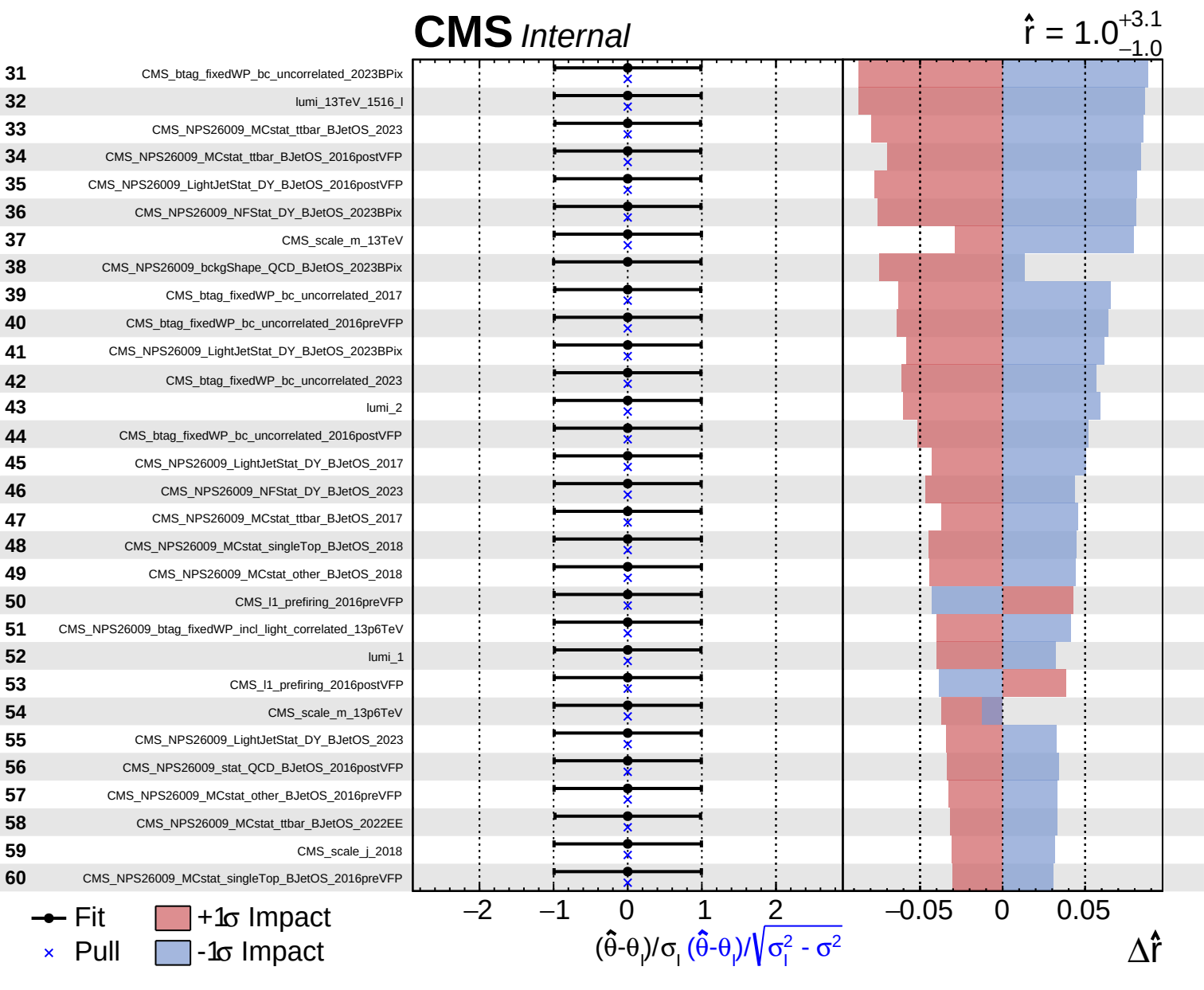
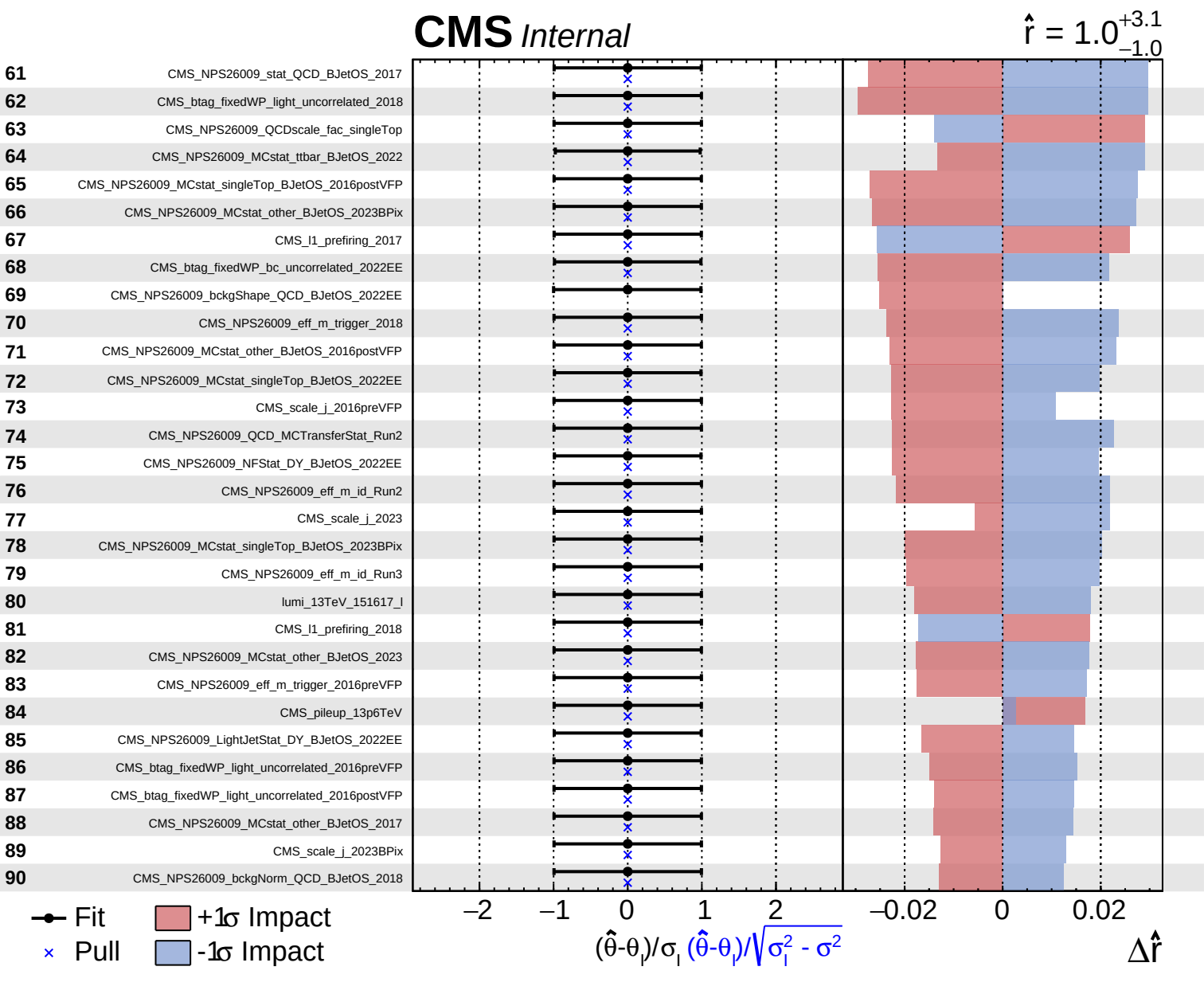
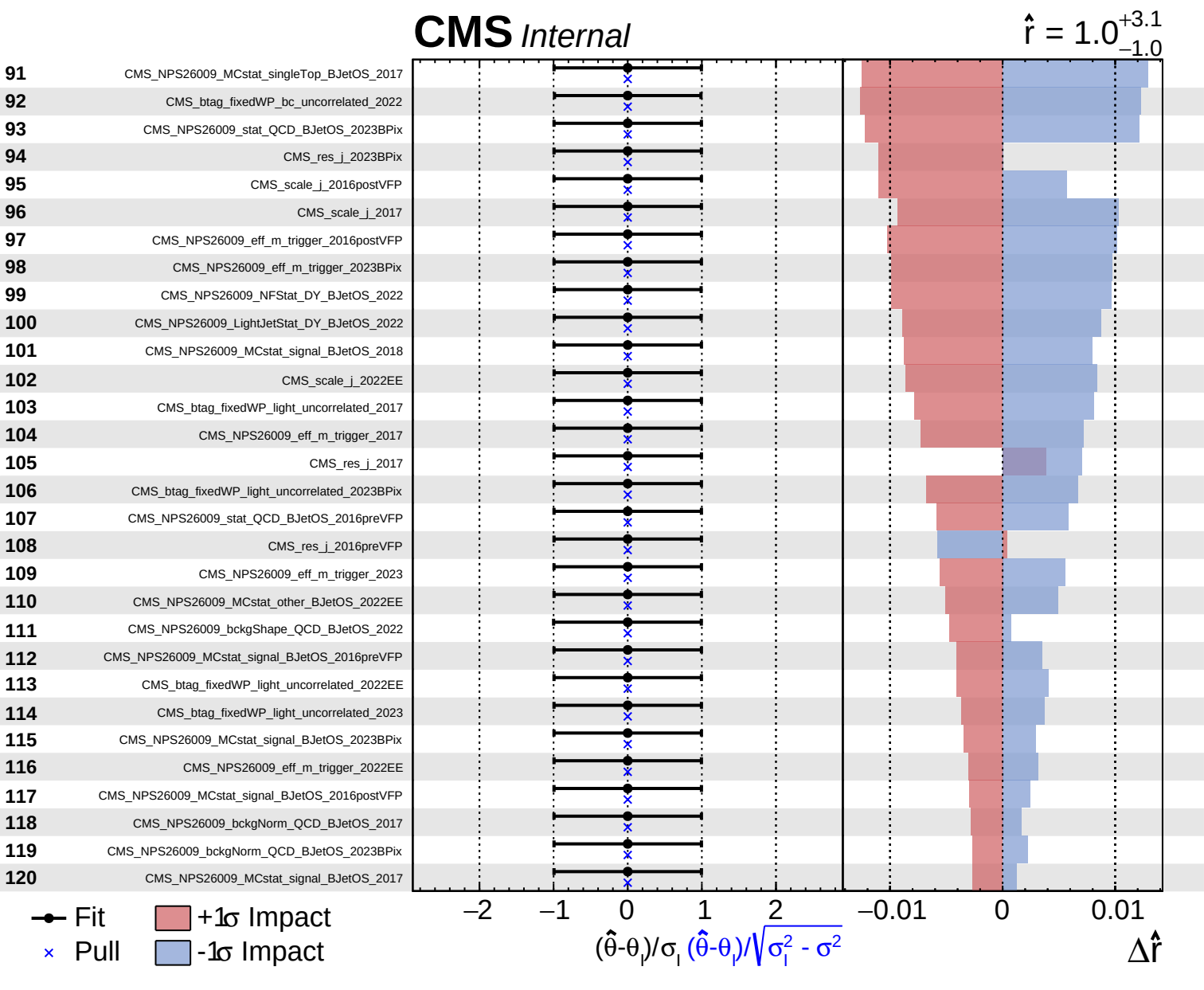
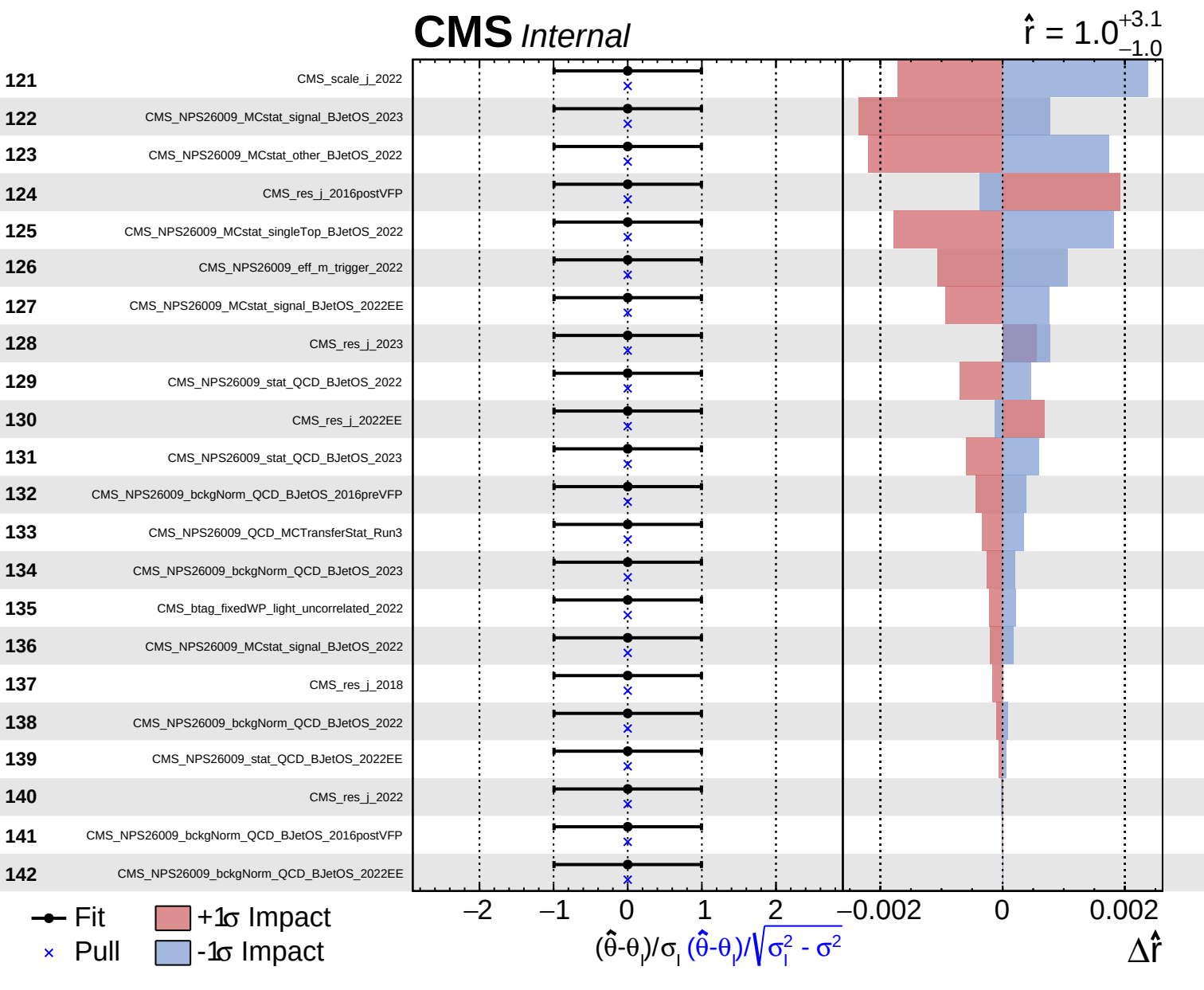

CMS Internal
$\hat{r} = 1.0^{+3.1}_{-1.0}$
1 QCDscale_ren_ttbar
2 QCDscale_fac_ttbar
3 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
4 CMS_NPS26009_topmass_ttbar_BJetOS
5 pdf_alphas
6 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
7 CMS_NPS26009_NFStat_DY_BJetOS_2018
8 CMS_NPS26009_pdf_NNPDF31
9 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
10 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
11 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
12 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
13 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
14 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
15 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
16 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
17 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
18 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
19 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
20 CMS_NPS26009_NFStat_DY_BJetOS_2017
21 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
22 CMS_btag_fixedWP_bc_uncorrelated_2018
23 lumi_13TeV_15161718_I
24 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
25 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
26 CMS_NPS26009_stat_QCD_BJetOS_2018
27 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
28 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
29 CMS_pileup_13TeV
30 CMS_NPS26009_QCDscale_ren_singleTop
−2 −1 0 1 2
−2 −1 0 1 2
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+3.1}_{-1.0}$
31 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
32 lumi_13TeV_1516_I
33 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
34 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
35 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
36 CMS_NPS26009_NFStat_DY_BJetOS_2023BPix
37 CMS_scale_m_13TeV
38 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
39 CMS_btag_fixedWP_bc_uncorrelated_2017
40 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
41 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
42 CMS_btag_fixedWP_bc_uncorrelated_2023
43 lumi_2
44 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
45 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
46 CMS_NPS26009_NFStat_DY_BJetOS_2023
47 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
48 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
49 CMS_NPS26009_MCstat_other_BJetOS_2018
50 CMS_l1_prefiring_2016preVFP
51 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
52 lumi_1
53 CMS_l1_prefiring_2016postVFP
54 CMS_scale_m_13p6TeV
55 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
56 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
57 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
58 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
59 CMS_scale_j_2018
60 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.05
0
0.05
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+3.1}_{-1.0}$
61 CMS_NPS26009_stat_QCD_BJetOS_2017
62 CMS_btag_fixedWP_light_uncorrelated_2018
63 CMS_NPS26009_QCDscale_fac_singleTop
64 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
65 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
66 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
67 CMS_l1_prefiring_2017
68 CMS_btag_fixedWP_bc_uncorrelated_2022EE
69 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
70 CMS_NPS26009_eff_m_trigger_2018
71 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
72 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
73 CMS_scale_j_2016preVFP
74 CMS_NPS26009_QCD_MCTransferStat_Run2
75 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
76 CMS_NPS26009_eff_m_id_Run2
77 CMS_scale_j_2023
78 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
79 CMS_NPS26009_eff_m_id_Run3
80 lumi_13TeV_151617_l
81 CMS_l1_prefiring_2018
82 CMS_NPS26009_MCstat_other_BJetOS_2023
83 CMS_NPS26009_eff_m_trigger_2016preVFP
84 CMS_pileup_13p6TeV
85 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
86 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
87 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
88 CMS_NPS26009_MCstat_other_BJetOS_2017
89 CMS_scale_j_2023BPix
90 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.02
0
0.02
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+3.1}_{-1.0}$
91 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
92 CMS_btag_fixedWP_bc_uncorrelated_2022
93 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
94 CMS_res_j_2023BPix
95 CMS_scale_j_2016postVFP
96 CMS_scale_j_2017
97 CMS_NPS26009_eff_m_trigger_2016postVFP
98 CMS_NPS26009_eff_m_trigger_2023BPix
99 CMS_NPS26009_NFStat_DY_BJetOS_2022
100 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
101 CMS_NPS26009_MCstat_signal_BJetOS_2018
102 CMS_scale_j_2022EE
103 CMS_btag_fixedWP_light_uncorrelated_2017
104 CMS_NPS26009_eff_m_trigger_2017
105 CMS_res_j_2017
106 CMS_btag_fixedWP_light_uncorrelated_2023BPix
107 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
108 CMS_res_j_2016preVFP
109 CMS_NPS26009_eff_m_trigger_2023
110 CMS_NPS26009_MCstat_other_BJetOS_2022EE
111 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
112 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
113 CMS_btag_fixedWP_light_uncorrelated_2022EE
114 CMS_btag_fixedWP_light_uncorrelated_2023
115 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
116 CMS_NPS26009_eff_m_trigger_2022EE
117 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
118 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
119 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
120 CMS_NPS26009_MCstat_signal_BJetOS_2017
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.01
0
0.01
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+3.1}_{-1.0}$
121 CMS_scale_j_2022
122 CMS_NPS26009_MCstat_signal_BJetOS_2023
123 CMS_NPS26009_MCstat_other_BJetOS_2022
124 CMS_res_j_2016postVFP
125 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
126 CMS_NPS26009_eff_m_trigger_2022
127 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
128 CMS_res_j_2023
129 CMS_NPS26009_stat_QCD_BJetOS_2022
130 CMS_res_j_2022EE
131 CMS_NPS26009_stat_QCD_BJetOS_2023
132 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
133 CMS_NPS26009_QCD_MCTransferStat_Run3
134 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
135 CMS_btag_fixedWP_light_uncorrelated_2022
136 CMS_NPS26009_MCstat_signal_BJetOS_2022
137 CMS_res_j_2018
138 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
139 CMS_NPS26009_stat_QCD_BJetOS_2022EE
140 CMS_res_j_2022
141 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
142 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.002
0
0.002
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$